

Tensor Network techniques for differential equations

QTADE School, Bilbao

Raghavendra Dheeraj Peddinti

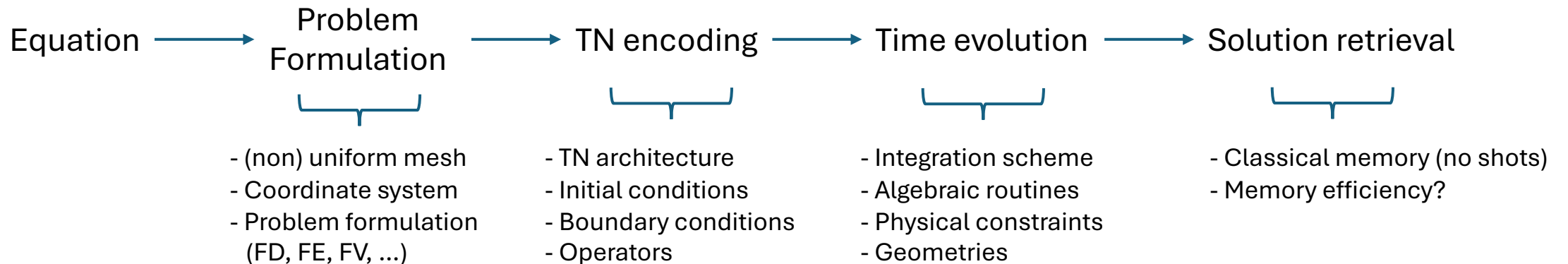
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Quantum Research Center, TII, Abu Dhabi

TUHH, Hamburg

What the lectures are about – high level

Use Tensor Networks (TNs) to exploit the low-rank structure of solutions and operators defining PDEs (no quantum)



Rank / Bond dimension: χ

Structure of the lectures

Today:

- Part I: Representing fields and operators (Stefano)
- Part II: Linear algebraic routines (Raghavendra)

Tomorrow afternoon:

... with Tensor Networks

- Part III: The forward problem: time evolution (Stefano)
- Part IV: The inverse problem: data-driven methods (Raghavendra)

Representing fields and operators

Tensors and Penrose notation

MPS or TT via SVD

Historical perspective on TNs

Quantics TT: functions and operators

Heat equation (explicit)

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Tensors

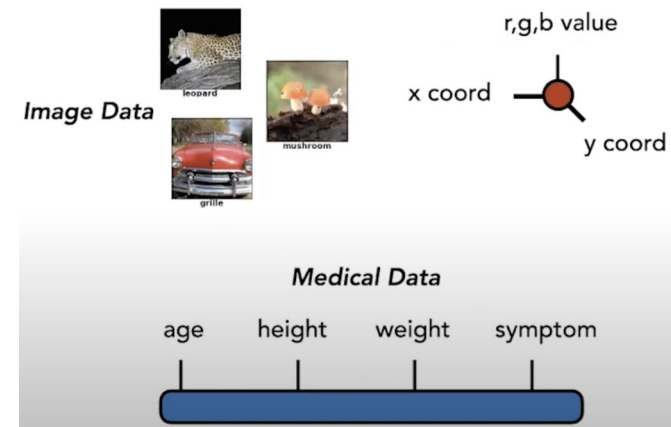
What is a Tensor?

Generalization of a vector or matrix:
Tensor = multi-dimensional array

vector	$v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$	$v_1 = 2$ $v_2 = 3$
matrix	$M = \begin{bmatrix} 5 & 7 \\ 8 & 9 \end{bmatrix}$	$M_{11} = 5$ $M_{12} = 7$
order-3 tensor	$T = \begin{bmatrix} 3 & \begin{bmatrix} 5 & 4 \end{bmatrix} & 7 \\ 1 & \begin{bmatrix} 3 & 2 \end{bmatrix} & 5 \end{bmatrix}$	$T_{111} = 3$ $T_{112} = 5$

Where do tensors occur?

Multi-dimensional data (image,
medicine...), discretization of functions,
many-body quantum wave-functions.



Tensor elements

Tensors beyond a few indices become **exponentially** costly to store and manipulate

$$T_{n_1 n_2 n_3 n_4 n_5 n_6}, \quad n_j = 1, 2, \dots, 10$$


10⁶ entries

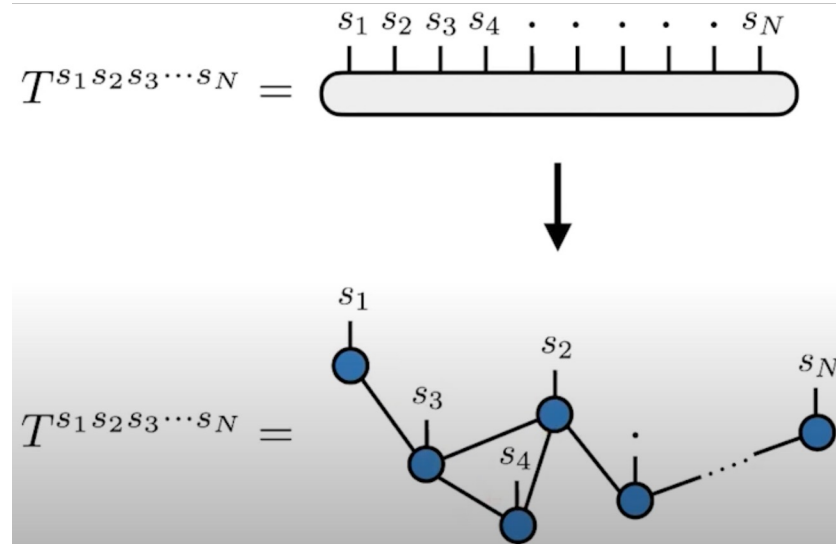
N indices -> entries **exponential** in N

Curse of
dimensionality

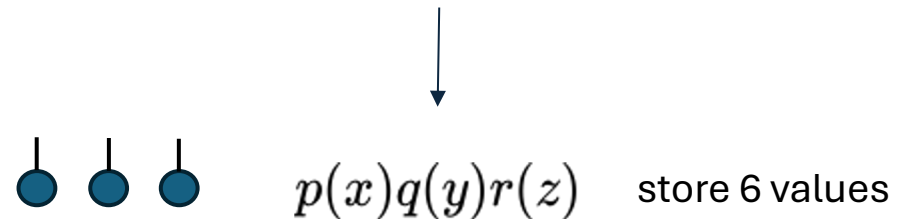
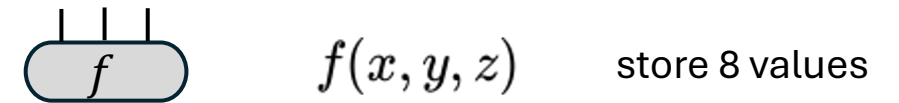
Structured tensors

TNs give a way to break the curse

N-index (order) tensor -> shape with N lines



Example: $x, y, z \in \{0,1\}$



$$f(0,1,0) = \begin{array}{c} 0 \quad 1 \quad 0 \\ | \quad | \quad | \\ \text{---} f \text{---} \end{array}$$

Tensor Diagram Notation (Roger Pensrose 1971)

Low-order tensor examples



vector: v_j

matrix: M_{ij}

order – 3 tensor: T_{ijk}

- No ambiguity
- names of indices can be omitted
- Clear computational cost (next slide)
- Covariant/contravariant indices by ingoing/outgoing arrow
- Special tensors -> diff. shapes (isometries are triangles)

- **Joined legs** — sum over the shared index
- **Closed loop** — no free legs, so a number (trace)

$$\text{Diagram} \longleftrightarrow \sum_j M_{ij} v_j$$

$$\text{Diagram} \longleftrightarrow A_{ij} B_{ji} = \text{Tr}[AB]$$

Tensor Diagram Notation <https://tensornetwork.org>

Examples

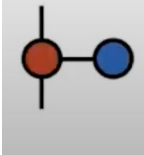

$$\sum_j v_j w_j$$

Vector inner prod



$$\sum_j M_{jj}$$

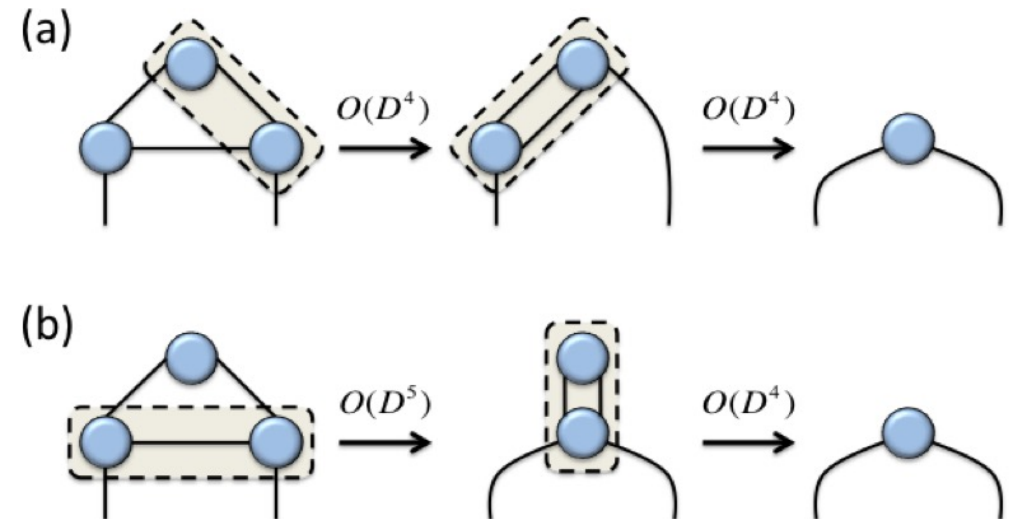
Matrix trace



$$\sum_j T_{ijk} v_j$$

Order-3 tensor
contracted
with a vector

Computational cost - explicit



Finding the optimal order for a *general* network is NP-hard; for a chain it is obvious.

Representing fields and operators

Tensors and Penrose notation

MPS or TT via SVD

Historical perspective on TNs

Quantics TT: functions and operators

Heat equation (explicit)

MPS or TT via SVD

Exploiting the low-rank structure of a matrix

Problem: cannot store/manipulate tensors with N indices

Ways out:

- Sparsity (when applicable)
- Sampling
- **Low-rank structure**

Uncovering low-rank structure for matrices is straightforward: **SVD**

$$M = U S V^\dagger \quad \begin{matrix} U^\dagger U = \mathbb{I}_r \\ V^\dagger V = \mathbb{I}_r \end{matrix}$$
$$= A B \quad \begin{matrix} A = U\sqrt{S} \\ B = \sqrt{S}V^\dagger \end{matrix}$$

runs over r values, $\text{rank}(M) = r$

S diagonal contains the singular values:

- Real
- Non-negative
- Decreasing ordered

Truncating the SVs gives optimal L_2 approximation

$$M \approx \tilde{U} \tilde{S} \tilde{V}^\dagger$$
$$= \tilde{A} \tilde{B}$$

MPS or TT via SVD

Example

Given a rectangular (4x3) matrix M

$$M = \begin{bmatrix} 0.435839 & 0.223707 & 0.10 \\ 0.435839 & 0.223707 & -0.10 \\ 0.223707 & 0.435839 & 0.10 \\ 0.223707 & 0.435839 & -0.10 \end{bmatrix}$$

can factorize as

$$\begin{bmatrix} 1/2 & -1/2 & 1/2 \\ 1/2 & -1/2 & -1/2 \\ 1/2 & 1/2 & 1/2 \\ 1/2 & 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 0.933 & 0 & 0 \\ 0 & 0.300 & 0 \\ 0 & 0 & 0.200 \end{bmatrix} \begin{bmatrix} 0.707107 & 0.707107 & 0 \\ -0.707107 & 0.707107 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$U \quad S \quad V^\dagger$

Matrices U and V have orthonormal columns

$$U^\dagger U = \mathbb{I}_r$$

$$V^\dagger V = \mathbb{I}_r$$

Truncation

$$M_2 = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ 1/2 & -1/2 & -1/2 \\ 1/2 & 1/2 & 1/2 \\ 1/2 & 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 0.933 & 0 & 0 \\ 0 & 0.300 & 0 \\ 0 & 0 & \text{X} \end{bmatrix} \begin{bmatrix} 0.707107 & 0.707107 & 0 \\ -0.707107 & 0.707107 & 0 \\ \text{X} & \text{X} & \text{X} \end{bmatrix}$$

$$\|M_2 - M\|^2 = (0.2)^2$$

$$M_3 = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ 1/2 & -1/2 & -1/2 \\ 1/2 & 1/2 & 1/2 \\ 1/2 & 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 0.933 & 0 & 0 \\ 0 & \text{X} & 0 \\ 0 & 0 & \text{X} \end{bmatrix} \begin{bmatrix} 0.707107 & 0.707107 & 0 \\ \text{X} & \text{X} & \text{X} \\ \text{X} & \text{X} & \text{X} \end{bmatrix}$$

$$\|M_3 - M\|^2 = (0.2)^2 + (0.3)^2$$

MPS or TT via SVD

2 compression strategies for a matrix via SVD

Define a maximum allowed rank $:= \chi$



This defines the approximation error:

$$\epsilon = \sqrt{\sum_{j>\chi} \lambda_j^2}$$

Define a maximum allowed error $:= \epsilon$



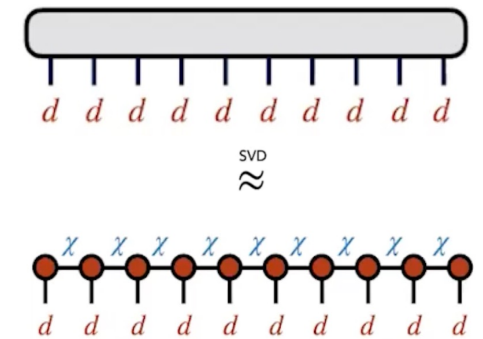
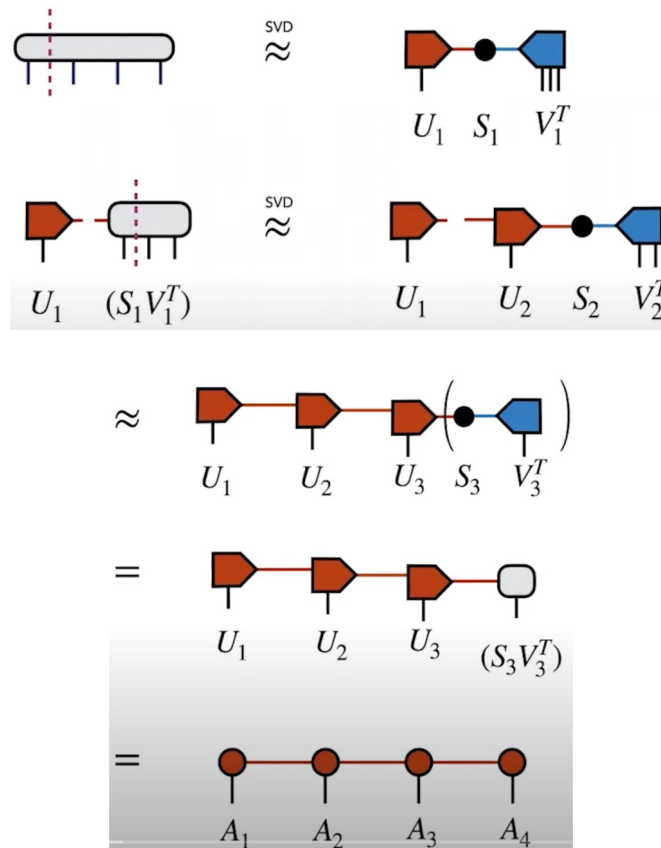
This defines the maximal χ such that:

$$\epsilon \geq \sqrt{\sum_{j>\chi} \lambda_j^2}$$

MPS or TT via SVD

Recursive SVD on a tensor

1. Reshape the tensor as a matrix
2. Apply SVD on the matrix
3. Glue together (multiply) SV^T
4. Sweep through the tensor



Matrix Product State

-
Tensor train

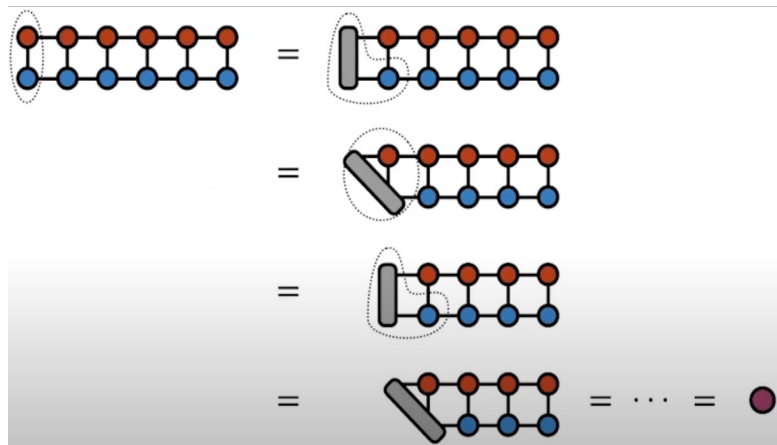
Compression:

$$d^N \longrightarrow N d \chi^2$$

MPS or TT via SVD – primitives

Efficient manipulations of MPS

- 1) Sum (χ is sub-additive)
- 2) Element-wise multiplication (recursive sketching alg.)
- 3) Perfect sample
- 4) Linear system solver (next lecture)
- 5) Inner product



$$\text{cost} \sim \chi^3$$

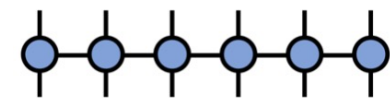
Inner product of two vectors
of a *trillion* entries each
(40 tensors with $d=2$):

10ms for $BD=100$

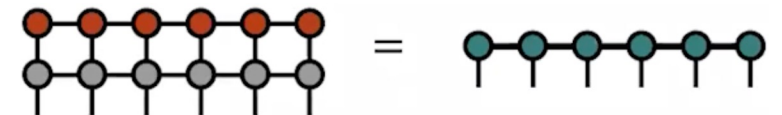
More advanced algorithms

- 1) Canonicalization (gauge freedom)
- 2) Belief propagation (gauging)
- 3) TCI (query T to sketch)
- 4) Multiscale Interpolative construction
- 5) Time evolution (TEBD, TDVP)

MPO / TT operator



MPO-MPS contraction



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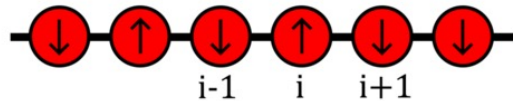
Historical perspective on TNs

Quantics TT: functions and operators

Heat equation (explicit)

Hystorical background

MPS have emerged in **quantum many-body physics** to study 1D spin chains



- DMRG as a variational algorithm in MPS manifold (S. White, 1992)
- Ground states of 1D local gapped Hamiltonians
- Entanglement-informed ansatz (area law)

Early Precursors in Physics (1940s–1980s)

- **Kramers and Wannier (1941)**: Introduced transfer matrices for the 2D Ising model, acting as early Matrix Product Operators.
- **Baxter (1968, 1981)**: Developed corner transfer matrices and used variational wave function precursors resembling MPS to calculate thermodynamic properties.
- **Fannes, Nachtergaele, and Werner (1992)**: Formally defined finitely correlated states in quantum spin chains, mathematically establishing what are now called MPS.  [arXiv.org +1](#)

Breakthrough via DMRG (1992–2000s)

- **White (1992)**: Invented the [Density Matrix Renormalization Group \(DMRG\)](#), a massive numerical breakthrough for 19th-century quantum lattice problems.
- **Ostlund and Rommer (1995)**: Realized that White's successful DMRG algorithm implicitly assumed the underlying quantum ground state was an MPS.
- **Vidal, Verstraete, and Cirac (2000s)**: Standardized MPS as a formal language for quantum entanglement, time evolution, and 1D many-body systems.  [arXiv +3](#)

Independent Discovery as Tensor Trains (2000s–Present)

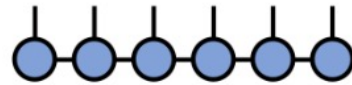
- **Oseledets and Tyrtysnikov (2009–2011)**: Independently rediscovered the exact same mathematical structure in numerical linear algebra, naming it the Tensor Train (TT) decomposition.
- **Applied Mathematics**: Brought MPS tools into data science, machine learning, and high-dimensional differential equations under the TT framework to manage massive datasets.  [arXiv.org +2](#)

Other Tensor Networks

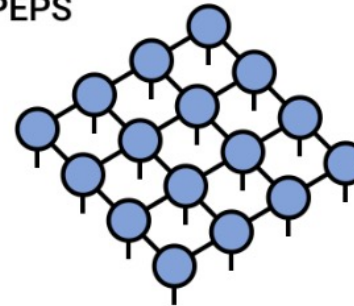
- Loopless $\rightarrow$ canonical form
 - No critical correlations (exp decay)
 - Easy manipulations (BD^{**3})
- (many known efficient algorithms and MPOs)**

Expressivity VS Computational Cost

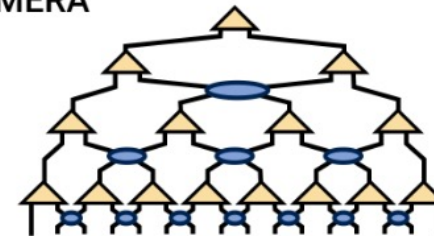
Matrix Product State /
Tensor Train



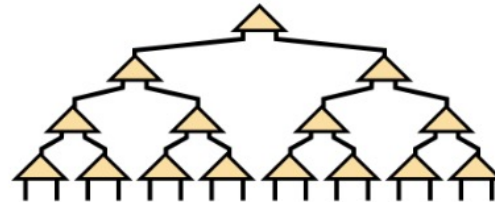
PEPS



MERA



Tree Tensor Network /
Hierarchical Tucker



- Loopy $\rightarrow$ no canonical form from single tensor operation
- Critical correlations
- Not easy manipulation (BD^{**8})

- Loopless $\rightarrow$ canonical form
- Critical correlations
- Ok-ish manipulations (BD^{**4})

Compressing multivariate functions with tree tensor networks

J Tindall, EM Stoudenmire, R Levy
arXiv preprint arXiv:2410.03572

Applications of TNs

- Functions of Continuous Variables (Quantum-inspired)
- Machine learning (compression of ML models/NN parametrization via TN)
- Probability and statistics (parametrizing statistical models)
- Quantum Physics (quantum simulation/error mitigation/process tensor)
- Out-of-equilibrium Statistical Physics
- Stochastic Chemical Kinetics

Libraries out there

- iTensor
- TeNPy
- TensorNetwork
- Quimb
- Cotengra (contraction optimization)

Hyper-optimized tensor network contraction
(J. Gray, S. Kourtis - 2021)

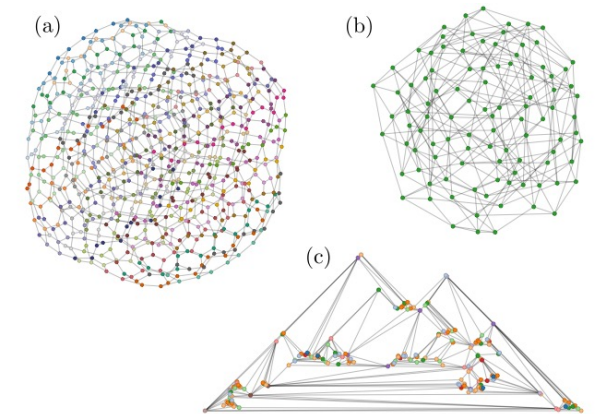


Figure 1: Sample tensor networks: (a) simplified network for a rectangular 7x7 qubit 1 + 40 + 1 depth random quantum circuit with 742 rank-3 tensors; (b) a random 5-regular network with 100 tensors, arising in, e.g., SAT problems; and (c) random planar network with 184 tensors, arising in, e.g., the statistical-mechanical evaluation of knot invariants.

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Quantics TT

Key idea: the tensor cores label spatial scales in a dyadic fashion

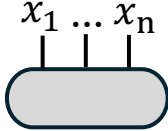
- The physical indices of the TT are labeling the domain discretization
- The values of the function are encoded in the contraction of the matrices ($\sim$ A.E.)

$\sim$ Amplitude encoding

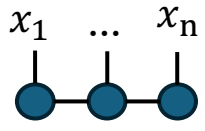
$$|f\rangle = \sum_{x_1, \dots, x_n \in \{0,1\}} f_{x_1, \dots, x_n} |x_1\rangle \otimes \dots \otimes |x_n\rangle$$

↓

order-n
tensor f

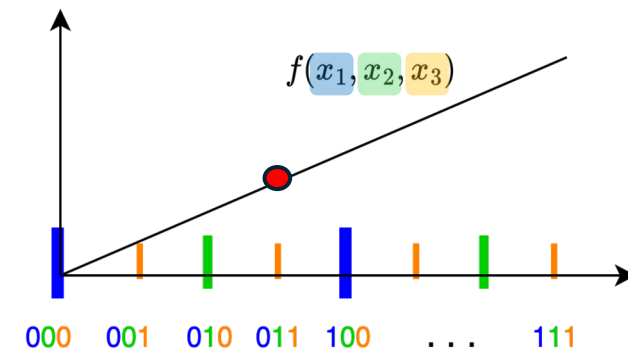


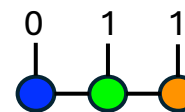
$\stackrel{\text{SVD}}{\approx}$



Mesh size scales $\exp$ in n
 BUT limited correlations (low χ) keep the memory cost linear

$f(x_1, x_2, x_3)$ with $x_i \in \{0,1\}$



$$f(0,1,1) =$$


Quantics TT

Key idea: the tensor cores label spatial scales in a dyadic fashion

- The physical indices of the TT are labeling the domain discretization
- The values of the function are encoded in the contraction of the matrices (A.E.)

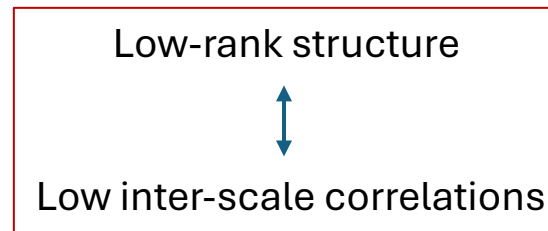
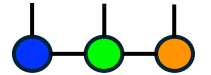
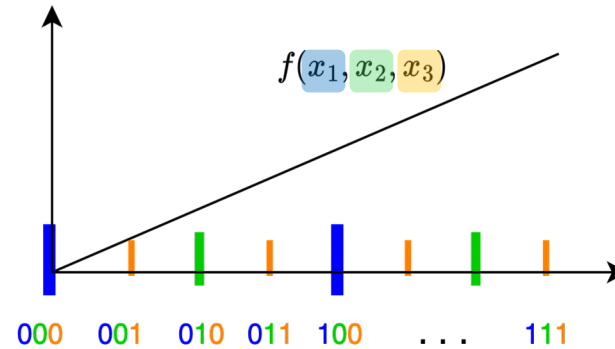
$f: [0,1) \rightarrow \mathbb{R}$ defined on $N = 2^n$ points

Binary indices for the discretized domain

$$f = f\left(\sum_{m=1}^n 2^{-m} x_m\right) = f\left(\frac{1}{2}x_1 + \frac{1}{4}x_2 + \dots + \frac{1}{2^n}x_n\right)$$

$$x_m \in \{0,1\}$$

x_1 : coarsest scale $\longrightarrow$ x_n : finest scale



Quantics TT

Analytical construction of 1D functions as QTT

Let's look at some simple 1D functions: rank is independent of the number of scales/qubits n

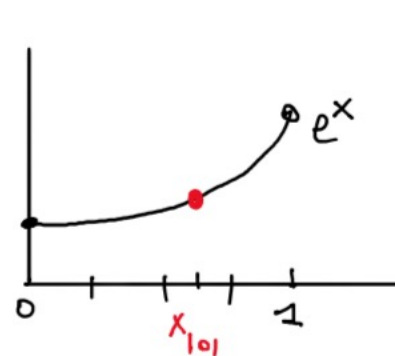
exp(x) $e^x = e^{x_1/2 + x_2/4 + \dots + x_n/2^n} = (e^{1/2})^{x_1} (e^{1/4})^{x_2} \dots (e^{1/2^n})^{x_n} = \begin{pmatrix} e^0 \\ e^{1/2} \end{pmatrix} \otimes \begin{pmatrix} e^0 \\ e^{1/4} \end{pmatrix} \otimes \dots \otimes \begin{pmatrix} e^0 \\ e^{1/2^n} \end{pmatrix}$

Rank = 1



sin(x), cos(x) Rank = 2

monomial: x^p Rank = p+1



$$e^{x_{101}} = \begin{pmatrix} 1 \\ e^{1/2} \end{pmatrix} \begin{pmatrix} 1 \\ e^{1/4} \end{pmatrix} \begin{pmatrix} 1 \\ e^{1/8} \end{pmatrix} = e^{5/8}$$

Quantics TT

Analytical construction of 1D functions as QTT

$$\stackrel{n=2}{\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \left[\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right]$$

$$\stackrel{n=3}{\begin{pmatrix} 1 \\ 2 \\ \vdots \\ 8 \end{pmatrix}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes [n=2] + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \left[[n=2] + 2^2 \begin{pmatrix} 1 \\ 1 \end{pmatrix}^{\otimes 2} \right]$$

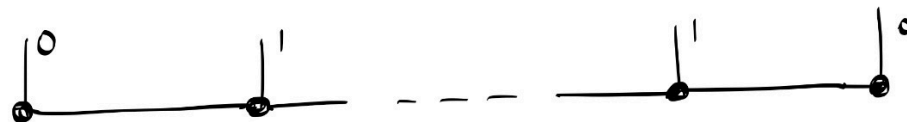
RECURSIVELY

$$\begin{pmatrix} 1 \\ 2 \\ \vdots \\ 2^n \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes [n-1] + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes 2^{n-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix}^{\otimes n-1} \rightarrow \begin{matrix} \chi=2 \text{ across} \\ \text{each bond} \end{matrix}$$

$$f(x) = x$$

Rank = 2

$$\Rightarrow \left[\begin{pmatrix} \cdot \\ \cdot \end{pmatrix} \begin{pmatrix} \cdot \\ \cdot \end{pmatrix} \right] \otimes \left[\begin{pmatrix} \cdot \\ \cdot \end{pmatrix} \begin{pmatrix} \cdot \\ \cdot \end{pmatrix} \right] \otimes \dots \otimes \left[\begin{pmatrix} \cdot \\ \cdot \end{pmatrix} \begin{pmatrix} \cdot \\ \cdot \end{pmatrix} \right] \otimes \left[\begin{pmatrix} \cdot \\ \cdot \end{pmatrix} \begin{pmatrix} \cdot \\ \cdot \end{pmatrix} \right]$$



Quantics TT

What functional class?

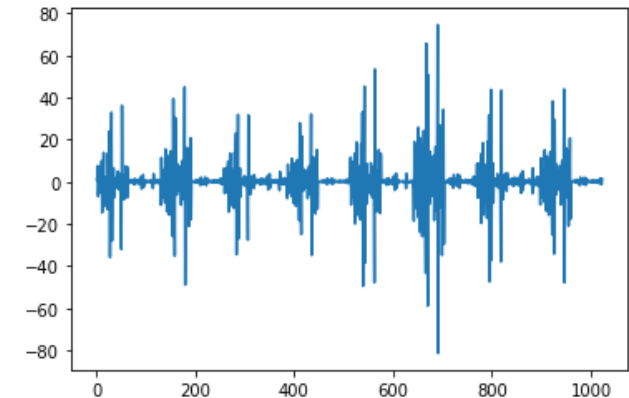
- 15. ***Approximation Theory of Tree Tensor Networks: Tensorized Univariate Functions***, Mazen Ali, Anthony Nouy, *Constr Approx*
- 16. ***Approximation Theory of Tree Tensor Networks: Tensorized Multivariate Functions***, Mazen Ali, Anthony Nouy, *arxiv:2101.11932*
- 17. ***Multiscale interpolative construction of quantized tensor trains***, Michael Lindsey, *arxiv:2311.12554*

TTn (QTT are subclass) achieve optimal approximation order for smooth functions (Sobolev or Besov spaces) of any smoothness order.

- Constructive proofs + efficient compression algs.
- Decaying ranks and rank bounds on smooth functions (band-limited) and functions with cusps or discontinuities.

BUT

Random MPS
with $\chi = 2$



Connection MERA-wavelet

[Submitted on 3 Feb 2016 (v1), last revised 16 Feb 2016 (this version, v2)]

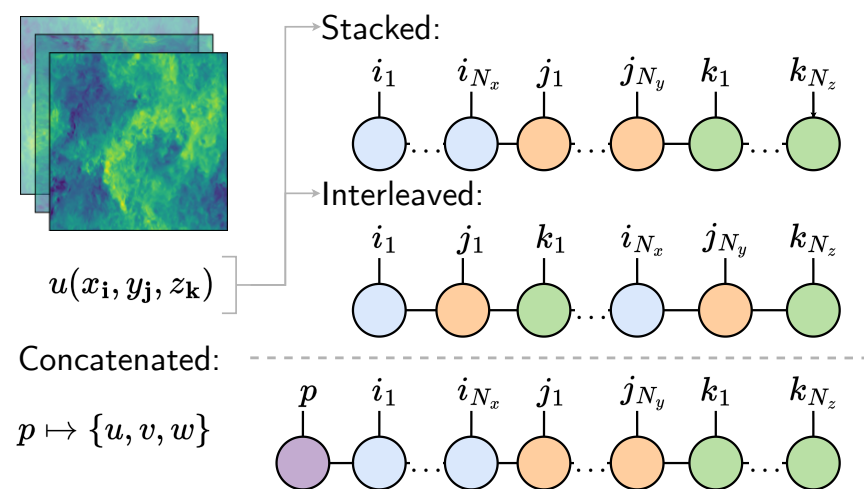
Entanglement renormalization and wavelets

Glen Evenbly, Steven R. White

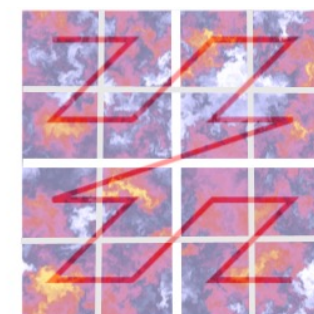
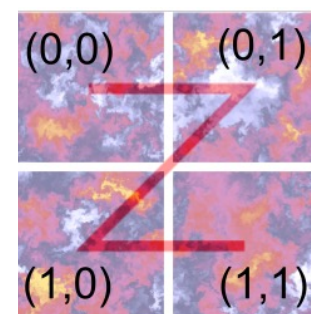
Quantics TT

Higher dimension: 2D, 3D, D+T

- We have the freedom to choose the bit ordering in higher dimesions
- Physically motivated (Heat equation later)



Interleaved (Morton/Z – ordering)



“Image compression and entanglement”
J.I. Latorre (2005)

Quantics TT

Analytical construction of finite difference operators as MPO

Laplace operator: $\chi = 3$

Low-Rank Explicit QTT Representation of the Laplace Operator and Its Inverse

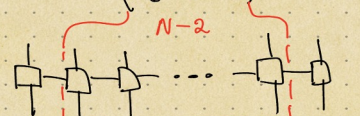
Authors: Vladimir A. Kazeev and Boris N. Khoromskij | AUTHORS INFO & AFFILIATIONS

<https://doi.org/10.1137/100820479>

Provable Low-Rank Tensor-Train Approximations in the Inverse of Large-Scale Structured Matrices

Chuanfu Xiao, Kejun Tang, Zhitao Zhu

Each entry is now a 2x2 matrix!

$$\Delta = (\cdot \cdot \cdot) \boxtimes \begin{pmatrix} \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \end{pmatrix} \boxtimes \begin{pmatrix} \vdots \\ \vdots \\ \vdots \end{pmatrix}$$


$$\Delta_8 = \begin{bmatrix} 2 & -1 & & & & & & \\ -1 & 2 & -1 & & & & & \\ & -1 & 2 & -1 & & & & \\ & & -1 & 2 & -1 & & & \\ & & & -1 & 2 & -1 & & \\ & \emptyset & & & -1 & 2 & -1 & \\ & & & & & -1 & 2 & -1 \\ & & & & & & -1 & 2 \end{bmatrix}_{8 \times 8}$$

define: $\mathbb{I}_2 \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $J \equiv \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$\Delta_2 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \mathbb{I}_2 - (J + J^T)$$

$$\Delta_4 = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \otimes \Delta_2 + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \otimes \Delta_2 - \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \otimes J^T - \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \otimes J$$

$$= \mathbb{I}_2 \otimes \Delta_2 - (J \otimes J^T + J^T \otimes J)$$

$$= \begin{pmatrix} \mathbb{I}_2 & -J & -J^T \end{pmatrix} \boxtimes \begin{pmatrix} \Delta_2 \\ J^T \\ J \end{pmatrix}$$

$$\Delta_{2^N} = \begin{pmatrix} \mathbb{I}_2 & -J^T & -J \end{pmatrix} \boxtimes \underbrace{\begin{pmatrix} \mathbb{I}_2 & -J & -J^T \\ 0 & 0 & J \\ 0 & J^T & J \end{pmatrix}}_{N-2 \text{ times}} \boxtimes \begin{pmatrix} \Delta_2 \\ J^T \\ J \end{pmatrix}$$

Remark: BCs can be included at small cost

Quantics TT

QFT “dequantization”, Laplace transform

Quantum Fourier transform has small entanglement

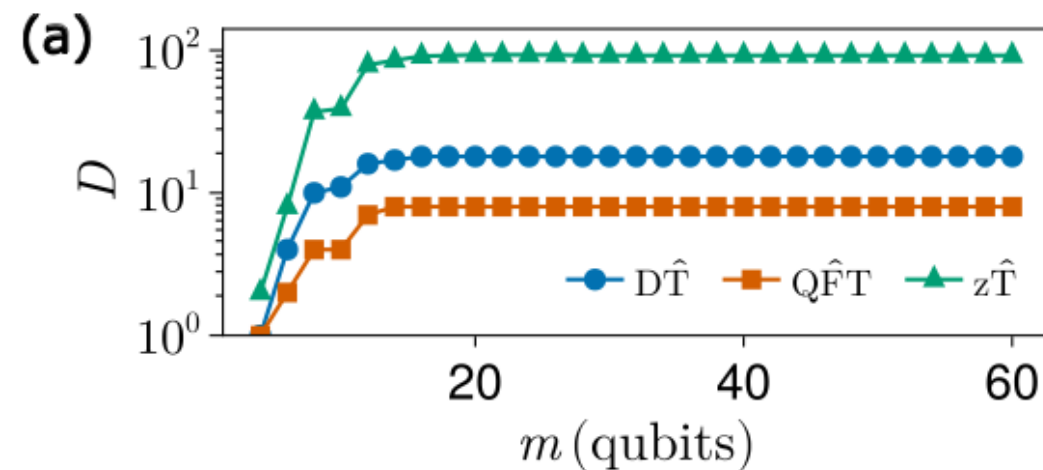
J Chen, EM Stoudenmire, SR White
PRX Quantum 4 (4), 040318

Direct interpolative construction of the discrete Fourier transform as a matrix product operator

J Chen, M Lindsey
Applied and Computational Harmonic Analysis, 101817

Quantum-inspired algorithms beyond unitary circuits: the laplace transform

N Jaseem, S Ramos-Calderer, D Wang, JI Latorre, D Poletti
arXiv preprint arXiv:2601.17724



Quantics TT

Quantum-inspired algorithms for
multivariate analysis:
from interpolation to partial differential
equations

JJ Garcia Ripoll, 2021

Problem	Algorithm	Type	Cost
Expected value	Monte Carlo	C	$\mathcal{O}(1/\varepsilon^2)$ [3]
"	Amplitude estimation	Q	$\mathcal{O}(1/\varepsilon)$ [3]
"	MPS	QI	$\mathcal{O}(N\chi^3 \log_2(1/\varepsilon))$ [cf. §4.3]
Fourier transform	QFT	Q	$\mathcal{O}(Nm \log(Nm))$ [17]
"	FFT	C	$\mathcal{O}(Nm2^{Nm})$ [18]
"	MPS QFT	QI	$\mathcal{O}(Nm \times \text{Simp}_{Nm})$ [cf. §5.1]
Interpolation	Linear ($k = 1$)	C	$\mathcal{O}(2^{Nm})$ [18]
"	MPS Linear ($k = 1$)	QI	$\sim \text{Simp}_{Nm}$ [cf. §5.2.1]
"	FFT	C	$\mathcal{O}(N(m+k)2^{N(m+k)})$ [cf. §5.2.2]
"	MPS QFT	QI	$\sim \text{QFT}_{N(m+k)}$ [cf. §5.2.2]
PDE Evolution	Finite differences	C	$\mathcal{O}(T_{\text{cgs}}2^{2Nm})$ [18]
"	MPS differences	QI	$\mathcal{O}(T_{\text{cgs}} \times \text{Simp}_{Nm})$ [cf. §5.4.1]
"	FFT method	C	$\mathcal{O}((Nm+1)2^{Nm})$ [18, 19]
"	MPS QFT	QI	$\sim 2 \times \text{QFT}_{N(m+k)}$ [cf. §5.4.2]
State construct	GR-like	Q	$\mathcal{O}(Nm\chi^2)$ [cf. §3.3]
"	Explicit wavefunction	C	$\mathcal{O}(2^{Nm})$
"	MPS	QI	$\mathcal{O}(T_{\text{steps}} \times \text{Simp}_{Nm})$ [e.g. §4.5]
MPS algorithms	Simplification (Simp_{Nm})	C	$\mathcal{O}(T_{\text{sweeps}} Nm 4d^3 \chi^3)$ [20]
"	Expected values	C	$\mathcal{O}(Nm \times 2d\chi^3)$ [20]
"	MPO-MPS product	C	$\mathcal{O}(Nm(d\chi\chi_f)^2)$ [20]

Representing fields and operators

Tensors and Penrose notation

MPS or TT via SVD

Historical perspective on TNs

Quantics TT: functions and operators

Heat equation (explicit)

Heat equation (explicit time stepping)

$$\partial_t u = \alpha \partial_x^2 u$$

$$x \in [0,1]$$

$$N = 2^n \text{ points}$$



- Initial condition u_0 as an MPS/QT
- Discrete laplacian L as an MPO of rank 3
- Diffusion number: $\alpha \Delta t / \Delta x^2$

Explicit Euler

$$u^{k+1} = (1 + \alpha \Delta t L) u^k$$



2 operations:
MPO-MPS product + rounding (χ, ϵ)

Cost: $\mathcal{O}(n\chi^3)$ per step

Memory: $\mathcal{O}(n\chi^2)$



Both are logarithmic in the mesh size N

Stability remark

$$\alpha \Delta t / \Delta x^2 \leq 1/2$$

But:

$$\Delta x = 2^{-n} \Rightarrow \Delta t \sim 4^{-n}$$

So:

step count grows exp,
but memory linear in n

Rank remark

The diffusive dynamics smooths the solution, so the rank should stay flat or fall!

But if we don't truncate at each step, the rank grows like $3^k \chi_0$

2 sources of error:

- Discretization
- Truncation

Heat equation (explicit time stepping)

- Notebook using quimb (interesting comparison between stacked and interleaved in 2D)
“01-heat-equation-qtmps-quimb.ipynb”
- 2601.03035 (Waintal, Huang, Groth) – pag. 60,61 (they use TCI + FFT-MPO)
“Who can compete with quantum computers?
Lecture notes on quantum inspired tensor networks computational techniques”

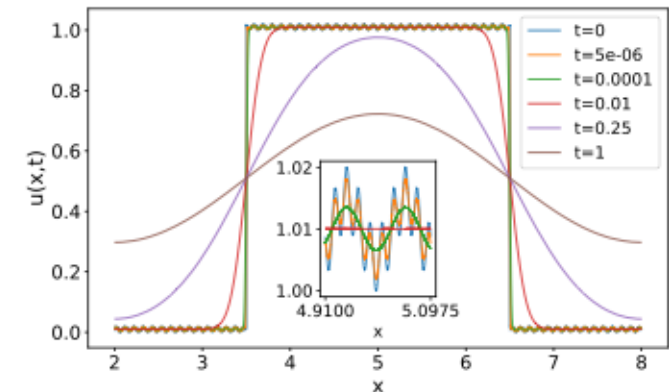
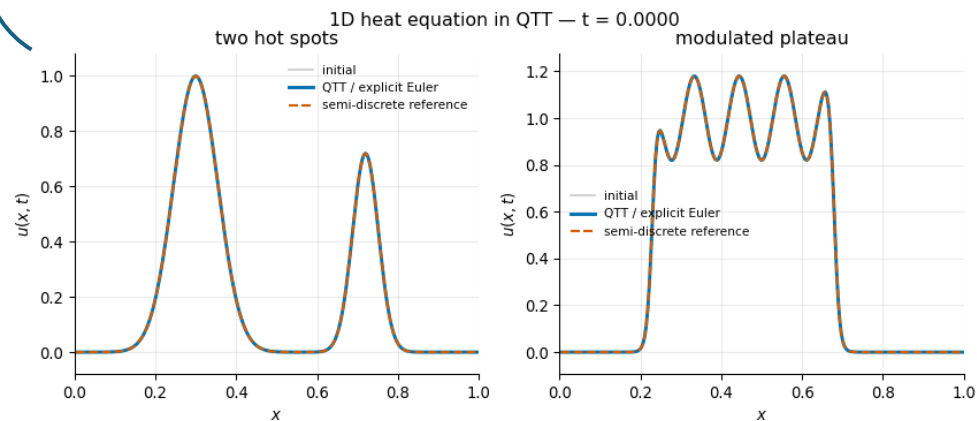


Figure 8: Illustration of using QTT to solve the heat equation (142) using quantics. The plot shows $u(x,t)$ versus x for different times, using an initial condition that contains Heaviside functions and rapidly oscillating terms: $u_0(x) = [1 + \cos(120x)\sin(180x)]/100 + \theta(x-7/2)[1 - \theta(x-13/2)]$. We used a 1D grid with 2^{30} points. The inset shows a zoom close to $x = 5$. (Adapted from [29]).